

SC - 205 Discrete Mathematics Project

Image Deblurring Using RL Algorithm



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1 Introduction

Discrete Mathematics is a field in arithmetic that handles the individual analytical makeups. It serves as a support for hypothetical Computer Science and involves miscellaneous attracting materials like Automata Theory, Cryptography, Graph Theory, photo deblurring, and many more. Discrete Mathematics ideas comprise a prerequisite to many centre ideas namely. artificial intelligence belief, prioritize communications network gauges, dictionary editor crafty, program growth, game development, dossier condensation, and the list is infinite. This project is freedom for us to investigate and mean a few of the abilities we have gained in our lectures. In today's occasion when the pubis is laboriously preoccupied with the new concerning details realm the amount of dossier namely being conceived and/or moved is pretty immense. There are differing habits at which point this dossieisbe insult controlled and intentional, skilled must be few habits to increase the effectiveness of news transfer the and to decrease its height and hold less thought. Hence, we understand utilizing this project as an aace for the unchanging.

2 Problem Statement

Problem : - Image deblurring using Richardson lucy algorithm

2.1 Brief Description of the Problem

Images are nowadays an integral part of our lives, whether in scientific applications or social networking and where there is an image, the concept of blurring might occur. Blurring is a major cause of image degradation and decreases the quality of an image. Blur occur due to the atmospheric commotion as well as the improper setting of a camera. Along with blur effects, noise also corrupts the captured image. Deblurring is the process of removing blurs and restoring the high-quality

latent image. Blur can be various types like Motion blur, Gaussian blur, Average blur, Defocus blur etc.

3 List of Algorithms

Here is a list of some commonly used algorithms for deblurring photograph,

- Richardson-Lucy Deconvolution
- Wiener filter
- Total variation Regularization
- Non blind Deconvolution
- Motion deblurring
- Deep learning based deblurring

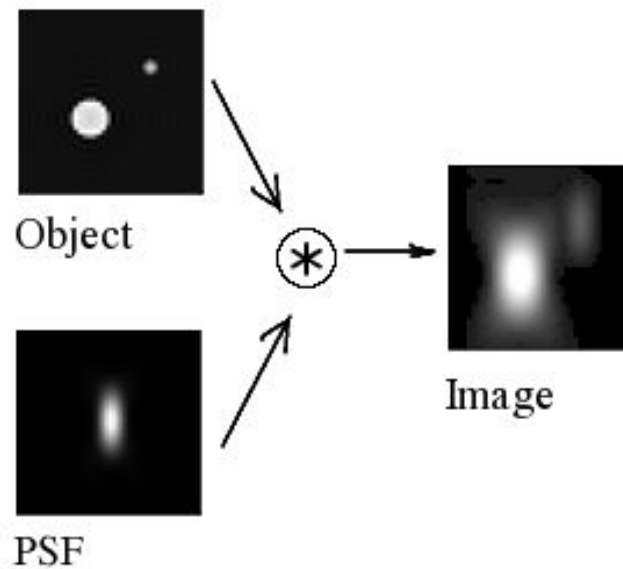
In our project we have tried to restore an image degraded by presence of noise by applying Richardson Lucy Algorithm. Given the noise function, one can use the Richardson-Lucy Algorithm to restore the degraded image. The algorithm is based on the principle of maximum likelihood estimation and assumes a mathematical model of the blurring process. It iteratively estimates the original image by refining an initial guess through a series of iterations. The algorithm assumes that the blurring process can be represented as a convolution between the original image and a point spread function (PSF) that describes the blurring effect.

4 Theory

The algorithm assumes a mathematical model of the degradation process, which is typically described as a convolution of the original image with a **Point Spread Function** (PSF). The PSF represents the blurring caused by factors such as optical aberrations, motion blur, or sensor characteristics. The goal of the Richardson-Lucy algorithm is to estimate the original image from the degraded image and the known

PSF. The algorithm starts with an initial estimate of the original image, often a blurred version of the observed image.

PSF(Point Spread Function) : - The point spread function (PSF) is a fundamental concept in imaging and optics. It refers to the response of an imaging system to a point source of light or a point object. The PSF describes how the system blurs or spreads out the light emitted from a point in the object space onto the image plane.



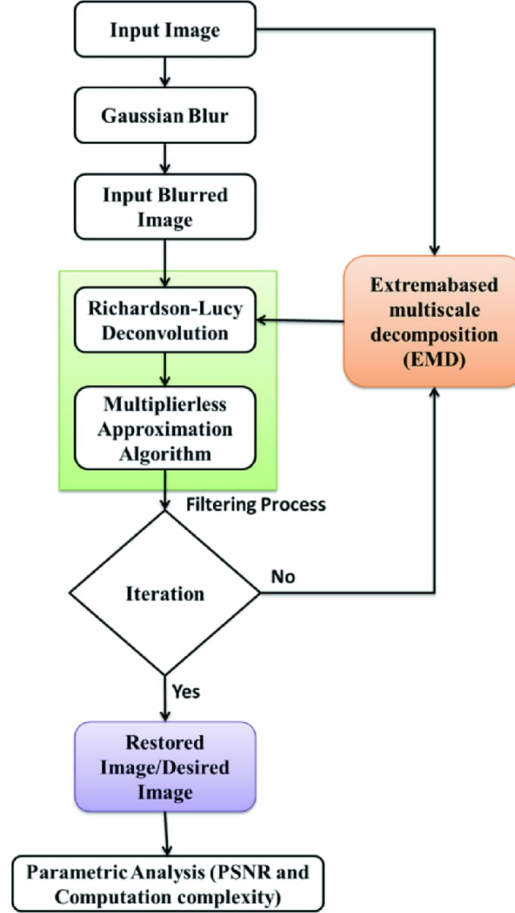
Example of the PSF

It then iteratively refines the estimate by alternating between two steps: the estimate update step and the estimate correction step.

1. **Estimate Update Step:** In this step, the current estimate is convolved with the PSF to obtain a blurred image. This blurred image is then divided element-wise by the observed degraded image, which essentially models the forward degradation process. The result is known as the restoration error.
2. **Estimate Correction Step:** In this step, the restoration error is convolved with the transposed PSF (also known as the point spread function adjoint or the Wiener filter) to obtain the correction factor. The current estimate is then multiplied element-wise by the

correction factor to update the estimate for the next iteration.

These two steps are repeated for a specified number of iterations or until a convergence criterion is met. The iterations gradually refine the estimate, aiming to minimize the difference between the estimated image convolved with the PSF and the observed degraded image. Starting from initializing the estimate of the original image and continuing until convergence or a specified number of iterations. The key steps such as convolution, division, computation of restoration error, convolution with the transposed PSF, multiplication with the correction factor, and updating the estimate are represented in the flowchart. Finally, the algorithm obtains the final estimate of the original image before stopping.



5 Algorithm

Let's denote the observed degraded image as G , the original image as F , the point spread function as H , and the noise as N . Each of these variables can be represented by discrete matrices or arrays. The Richardson-Lucy deconvolution algorithm in the discrete case can be written as:

- $\hat{F}(k)$ represents the estimated image at iteration k
- $\otimes$ denotes the discrete convolution operation
- G , F , and H are discrete matrices or arrays representing the observed degraded image, original image, and point spread function, respectively
- $\hat{H}(-1)$ represents the inverse of the matrix or array H

The initial estimate, $\hat{F}(0)$, can be an approximation of the original image or a constant value.

5.1 Step 1:-

Start with an initial estimate of the original image, denoted as $\hat{F}(0)$.

Initialize the iteration counter k to 0.

In the first step of the Richardson-Lucy algorithm for image deblurring, the estimated image "x" is initialized. This can be done by either setting "x" as the degraded or blurred image itself or by initializing it as a random noise image. The former assumes the blurred image is a reasonable approximation of the original, while the latter is useful when the degraded image contains significant noise or limited prior knowledge is available. The initialization step provides the starting point for the iterative process, and subsequent iterations refine the estimate based on the observed degraded image and the degradation model.

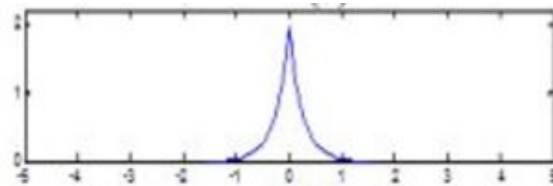


Original to Deblur

5.2 Step 2:-

In the second step of the Richardson-Lucy deconvolution algorithm, known as the **forward blurring** step.

At iteration k , we have the current estimate of the original image, $\hat{F}(k)$, which is obtained from the previous iteration or the initialization step. The PSF characterizes the blurring effect in the imaging system. It describes how a point source in the original image spreads out or blurs in the degraded image.



Graph of PSF

The convolution operation, denoted by the symbol " $\otimes$ ", is performed between the current estimate $\hat{F}(k)$ and the PSF H . Mathematically, it can be expressed

$$\text{Blurred_image} = \hat{F}(k) \otimes H$$

The output of the convolution operation is the estimated blurred image, denoted as `Blurred_image`. It represents the degradation of the original image due to blurring.

5.3 Step 3:-

In the third step of the Richardson-Lucy deconvolution algorithm, known as the **Estimation Update** step, the pixel-wise ratio between the observed degraded image, G , and the result of the forward blurring ($\hat{F}(k) \otimes H$) is computed. This ratio represents an estimation of the original image divided by the blurred estimate, which is adjusted by the observed image.

The forward blurring operation, performed in the previous step, produces the estimated blurred image ($\hat{F}(k) \otimes H$). This represents the degraded version of the original image obtained by convolving the current estimate with the PSF.

The estimation update is computed by taking the pixel-wise ratio between the observed degraded image, G , and the forward blurring result ($\hat{F}(k) \otimes H$). This can be expressed mathematically as:

$$\text{Estimation_update} = G / (\hat{F}(k) \otimes H)$$

Each pixel in the estimation update represents an estimate of the original pixel value, adjusted based on the observed image and the degradation model represented by the forward blurring operation.

The output of the estimation update step is the estimation update image, denoted as `Estimation_update`. It provides an updated estimate of the original image by considering the degradation and the observed degraded image.

5.4 Step 4:-

The fourth step in the Richardson-Lucy deconvolution algorithm for image deblurring is the Inverse Blurring.

Perform a convolution operation on the estimation update obtained in the previous step with the spatially reversed PSF, denoted as $H^*(-1)$.

This step aims to restore the image by counteracting the blurring effect. The estimation update obtained in the previous step, which represents an adjustment to the blurred estimate based on the observed degraded image, is convolved with the spatially reversed point spread function (PSF), denoted as $H^*(-1)$. The spatial reversal of the PSF is necessary to counteract the blurring effect introduced in the forward blurring step.

Convolution is a mathematical operation that combines two functions to produce a third function. In the context of image processing, convolution is used to describe the blurring effect caused by the PSF. The convolution operation can be defined as follows:

$$(F \otimes H)(x, y) = \sum [\sum (F(i, j) * H(x-i, y-j))]$$

The inverse blurring step is crucial for enhancing the sharpness and recovering fine details in the image. By applying the spatially reversed PSF, the algorithm attempts to undo the blurring process and restore the original image as much as possible.

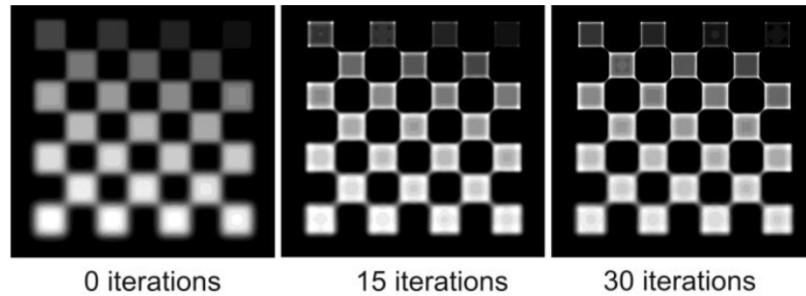
5.5 Step 5:-

The fifth step involves updating the estimate of the original image based on the result of the inverse blurring operation. This step aims to refine the estimate for the next iteration.

- Update:
 - Multiply the result of the inverse blurring, denoted as $I^*(k)$, with the current estimate of the original image, denoted as $F^*(k)$.
 - the multiplication is pixel-wise, meaning that each pixel value in $F^*(k)$ is multiplied with the corresponding pixel value in $I^*(k)$.
- Increment Iteration Counter:
 - Increment the iteration counter k by 1 to prepare for the next iteration.

The update step plays a crucial role in refining the estimate of the original image. By multiplying the inverse blurring result with the current estimate.

The iteration counter is incremented to keep track of the number of iterations performed. This counter helps control the convergence of the algorithm and ensures that the process terminates after a certain number of iterations or when a convergence criterion is met.



5.6 Step 6:-

In this step repeat steps 2-5 until a convergence criterion is met, such as a maximum number of iterations reached or the change between iterations falls below a threshold.

6 Code

```
1  #include <opencv2/opencv.hpp>
2  using namespace cv;
3
4  Mat richardsonLucyDeconvolution(const Mat& blurryImage, const Mat& psf, int iterations)
5  {
6      Mat currentEstimate = blurryImage.clone();
7      Mat psfFlipped;
8      flip(psf, psfFlipped, -1);
9
10     for (int i = 0; i < iterations; ++i)
11     {
12         Mat forwardBlurred;
13         filter2D(currentEstimate, forwardBlurred, -1, psf);
14
15         Mat estimationUpdate = blurryImage.mul(forwardBlurred);
16         filter2D(estimationUpdate, estimationUpdate, -1, psfFlipped);
17         divide(forwardBlurred, estimationUpdate, estimationUpdate, 1e-5);
18
19         currentEstimate = currentEstimate.mul(estimationUpdate);
20     }
21     return currentEstimate;
22 }
23 int main()
24 {
25     // Load the blurry image
26     Mat blurryImage = imread("blurry_image.png", IMREAD_GRAYSCALE);
27
28     // Define the point spread function (PSF)
29     Mat psf = ...; // Create or load the PSF
30
31     // Perform Richardson-Lucy deconvolution
32     int iterations = 10;
33     Mat deblurredImage = richardsonLucyDeconvolution(blurryImage, psf, iterations);
34
35     // Display the deblurred image
36     namedWindow("Deblurred Image", WINDOW_NORMAL);
37     imshow("Deblurred Image", deblurredImage);
38     waitKey(0);
39
40     return 0;
41 }
```

7 Real Life Application

- **Astronomical Imaging:** In astronomy, images captured by telescopes are often affected by atmospheric turbulence and instrumental blurring. This algorithm can be used to enhance the quality of astronomical images by deconvolving the effects of blurring and restoring fine details.
- **Medical Imaging:** Medical imaging techniques such as magnetic resonance imaging (MRI) and positron emission tomography (PET) can suffer from blurring caused by motion, limited resolution, or the imaging process itself. This algorithm can be applied to improve the clarity and resolution of medical images.
- **Microscopy:** Microscopic images can be affected by various factors such as diffraction, aberrations, and noise, leading to blurred and degraded images. Using this algorithm improve quality of the image.
- **Forensic Image Analysis:** In forensic investigations, images obtained from surveillance cameras or crime scene photography may suffer from blurring or noise. The Richardson-Lucy algorithm can be utilized to enhance these images, revealing crucial details and assisting in the identification and analysis of evidence.

8 Bibliography

References

[1] **Image Deblurring: Algorithms and Techniques** by Subhasis Chaudhuri - This book provides a comprehensive overview of image deblurring algorithms and techniques, covering both classical and modern approaches.

[2] Learn about Richardson-Lucy algorithm:

Wikipedia : <https://en.wikipedia.org/wiki/Talk:Richardson>

MathWork : <https://www.mathworks.com/>

